

JORGE URDANETA

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EDUCATION

Bachelor of Science in Mechanical Engineering

The University of Texas at Austin

Expected Graduation: Dec 2026

GPA: 3.2

Associate in Science in Mechanical Engineering, *Graduated with Honors*

Midland College

Graduated: May 2023

GPA: 3.9

TECHNICAL EXPERIENCE

Boeing Commercial Airplanes – 777/777X Program

Equipment Engineering Intern (Midbodies Future Builds)

May 2026 - Present

- Modernized 3 metrology kiosks via Windows 7 →10 migration, restoring the production metrology software stack
- Deployed and validated metrology automation networks linking Leica lasers, PCs, and SPM tablets for control/data transfer
- Evaluated UPS architectures and designed PC mounting bracket, reducing data loss and corruption risk by 90% in metrology
- Redesigned metrology WinCC HMI workflows, standardizing operator interactions and improving process consistency
- Planned end-to-end metrology test bench for next-gen laser systems, automating XML test procedures and SDK integration

Manufacturing Engineering Intern (Systems and Electrical Installation)

June 2025 - August 2025

- Restructured gyro battery housing in CATIA with integrated features and simplified wiring, cutting system footprint ~50%
- Engineered HUD inspection tool with GD&T to raise deviation sensitivity by 75% and catch > .015" hard-point shifts
- Prototyped new economy seat tray with features to fully prevent drink spills and secure phone in 3-day design-a-thon
- Flagged 24 Boeing 777-9 aircraft with outdated pilot seat builds to drive rework efforts and align with program standards

ASCEND Texas — GoAero & X-Prize Wildfire Challenge

Mechanical Design Lead

Jan 2026 – May 2026

- Guided 5-person mechanical design subteam developing paramotor propulsion subsystem, overseeing CAD, design reviews, and structural interface integration between ATV chassis, cage, and propulsion system
- Directed iterative prototyping incorporating FEA, thrust reactions and CoG studies under limited manufacturer data
- Managed final integration using 80/20 frame and commercial paramotor cage, validating fit, structure, and assembly viability
- Engineered development of a remote-controlled braking system integrating ATV brakes with a linear actuator

Mechanical Design & Production Engineer

Sept 2024 – May 2025

- Oversaw the removal of the gasoline engine and installation of electronic motor to convert ATV into fully electric vehicle
- Originated avionics enclosure concept prioritizing airflow, accessibility, and EMI shielding for system reliability
- Designed and led fabrication of suspended motor mount using C-channels, optimizing alignment and prioritizing assembly
- Aligned drivetrain layout to meet spacing constraints for motor, batteries, and controller with upgrade potential

Mini-Bowling Table Game

Mar 2025

- Built and assembled a fully functional prototype including alley, gutters, pins, and a motor-actuated linkage mechanism
- Created a scoop attachment that mimics human wrist motion at the output joint of a four-bar linkage bowling mechanism
- Finalized manufacturing and assembly of components, ensuring alignment, tolerance control, and reliable performance

Mass Flow Meter Optimization System – Software Engineer

Apr 2023

- Implemented Python algorithms to improve measurement accuracy and efficiency, reducing error margins by over 10%
- Conducted root cause analysis on system inefficiencies, finding and implementing solutions to improve performance by 15%
- Collaborated with a cross-functional team to integrate process improvements and ensure reliable algorithm functionality

LEADERSHIP EXPERIENCE

ASCEND Texas — Program Manager

May 2025 — June 2026

- Led transition from ground ATV prototype to hybrid, paramotor-based, system, enabling remote payload delivery
- Developed an AppSheet project management platform to coordinate 25+ interdisciplinary engineers within different projects
- Monitored risks, blockers, and schedule deviations, escalating issues and driving resolution to keep development on track
- ASCEND's paramotor ATV hybrid air vehicle concept selected for GoAero & NASA Innovation Grant Award

Longhorn Energy Club – Vice-President of External Relations

Jun 2024 – May 2025

- Initiated partnerships with energy-related companies and organizations enhancing visibility and impact by +100%
- Showcased students' sustainable projects and innovative solutions to multiple companies in energy at our stellar events
- Organized panel series with HDR, FERVO, and Google experts on strategies for meeting data center energy demands

SKILLS & ACCOMPLISHMENTS

- **Software:** Python, WinCC, Finite Element Analysis (FEA), SolidWorks, OnShape, CATIA, Bambu Studio, ideaMaker
- **Hardware:** CNC Machining, Manual Milling, Weldment, Machining Processes, Prototyping, Sensor Integration
- **Awards:** 2nd place at Boeing BCA ONE Intern Design-a-thon, Hispanic Scholarship Fund Scholar, Nike Scholarship Finalist